Digital Circuit System

1. The number of 4-line to-16-line decoders required to make an 8-line to-256-line decoder is
   a) 16
   b) 17
   c) 32
   d) 64

2. The minimum number of bits required to represent positive numbers in the range of –1 to –9 using twos complement representation is
   a) 2
   b) 3
   c) 4
   d) 5

3. MOS devices are used for VLSI because of
   a) their higher propagation delay
   b) lower silicon chip area required
   c) availability of n-channel and p-channel devices
   d) availability of enhancement and depletion mode MOSFETs

4. Standard TTL is a
   a) current sink logic
   b) current source logic
   c) non-saturated bipolar logic
   d) unipolar logic

5. Minimum number of NAND gates required to implement (A+B)(C+D) is
   a) 6
   b) 7
   c) 8
   d) 9

6. Which of the following gate acts as controlled buffer
   a) OR gate
   b) AND gate
   c) Inverter
   d) Ex-OR gate

7. Which of the following operations is commutative but not associative?
   a) AND
   b) OR
   c) Ex-OR
   d) NAND

8. The NAND-NAND realization is equivalent to
   a) AND-NOT realization
   b) AND-OR realization
   c) OR-AND realization
   d) NOT-OR realization

9. Consider the expression: Z = A⊕B⊕C⊕……., where A, B, C, … are the input variables and Z is the output variable. Z will be logic 1 if
   a) an even number of input variables are 1
   b) an odd number of input variables are 1
   c) an even number of input variables are 0
   d) an odd number of input variables are 0

10. AB + A⊕B is equivalent to
   a) AND operation
   b) OR operation
   c) Ex-NOR operation
   d) NOR operation

11. Karnaugh map is used
   a) to minimize the number of FLIP-FLOPs in a digital circuit
   b) to minimize the number of gates only in a digital circuit
   c) to minimize the number of gates and the fan-in requirements of the gates in a digital circuit
   d) to design gates

12. A sequential circuit usually consists of
   a) only FLIP-FLOPs
   b) FLIP-FLOPs and gates
   c) Only gates
   d) None of the above

13. The code used for labeling the cells of the K-map is
   a) natural BCD
   b) hexadecimal
   c) Gray
   d) Octal

14. When 7-segment LED displays are employed to display numbers, zero blanking arrangement is used to blank out
   a) all the zeros
   b) all the leading zeros
   c) all the trailing zeros
   d) the zero in the most significant location

15. Race around condition occurs in a J-K FLIP-FLOP when
    a) both the inputs are 0
    b) both the inputs are 1
    c) the inputs are complementary
    d) any one of the above input combinations is present

16. A FLIP-FLOP has two outputs which are
    a) always zero
    b) always one
    c) always complementary
    d) in one of the above states

17. A ring counter consisting of 4 FLIP-FLOPs will have
    a) 4 states
    b) 8 states
    c) 16 states
    d) infinite states

18. A 40bit synchronous counter uses FLIP-FLOPs with propagation delay time of 25nsec. each. The maximum possible time required for change of state will be
    a) 25nsec
    b) 50nsec
    c) 75nsec
    d) 100nsec

19. A mod-2 counter followed by a mod-5 counter is
    a) a mod-5 counter only
    b) a decade counter
    c) a mod-7 counter
    d) any one of the above

20. A pulse-stratcher is same as a
    a) free-running multivibrator
    b) bistable multivibrator
    c) monostable multivibrator
    d) latch

21. FLIP-FLOPs can be used to make
    a) latches
    b) bounce-elimination switches
    c) registers
    d) all of the above

22. An A/D converter whose speed of conversion for a given analog input range is constant, is
    a) Successive-approximation A/D converter
    b) Parallel-comparator A/D converter

c) Counter ramp A/D converter
d) Dual-slope A/D converter

23. A charge coupled device is
a) a magnetic device
b) a bipolar semiconductor device
c) a MOS device
d) none of the above

24. A memory in which the contents get erased when power failure occurs is
a) RAM
b) EAROM
c) PROM
d) ROM

25. Address bus width of a memory of size 2048 x 8 is
a) 8
b) 9
c) 10
d) 11

26. A shift register is a
a) random access memory
b) sequentially accessed memory
c) read only memory
d) content addressable memory

27. A demultiplexer can be used to realize a
a) counter
b) shift-register
c) combinational circuit
d) display system

28. The logic circuitry in ALU is
a) entirely combinational
b) entirely sequential
c) combinational cum sequential
d) none of the above

29. BCD expresses each decimal digit as
a) a string of 2-bits
b) a byte
c) a string of 8-bits
d) a string of 4-bits

30. The boolean expression (A + C')(B' + C') simplifies to
    a) C'(A' + B)
    b) C' + A'B
    c) B'C' + AB'
    d) None of the above

31. The DSP based microcomputer differs from the conventional microprocessor based micorcomputer in the following way
    a) they follow Von Neumann model for the computer and have a single bus for program and data memory
    b) they follow Harvard model for the computer and have separate bus for program and data memory
    c) the instructions are microcoded and require multiple cycles for execution
    d) most of the instructions require only a single cycle for execution

32. The interrupt input of 8085 which has the lowest priority is
    a) TRAP
    b) INTR
    c) RST7.5
    d) RST5.5

33. The frequency of crystal connected at the clock input of 8085 is 4MHz. The period of one T state is
    a) 500nsec
    b) 250nsec
    c) 750nsec
    d) 1000nsec

34. The I/O address space that can be supported by 8085 directly is
    a) 256 bytes
    b) 1Kbytes
    c) 128bytes
    d) 512bytes

35. The advantage of the serial data communication schmes over the parallel data communication scheme are
    a) always provides asynchronous data transfer between transmitter and receiver
    b) cheaper
    c) larger speed of transmission
    d) no need for serial to parallel and parallel to serial conversion

36. The data transfer scheme in which the data transfer is to be done quickly is
    a) programmed I/O
    b) interrupt driven I/O
    c) direct memory transfer
    d) all of the above

37. The BIU of 8086
    a) decodes the instruction
    b) executes the instruction
    c) generates the timing signals
    d) fetches the instruction code

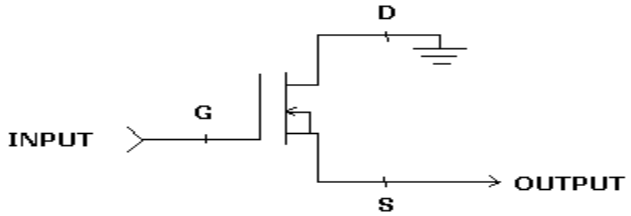
38. The least significant bits of a 16 bit number to be moved into a segment register should be
    a) 0000
    b) 1111
    c) can be any four bit number
    d) can't say

39. The maximum size of a single continuous stack segment is
    a) 64K
    b) 32K
    c) 16K
    d) 8K

40. The clock input of 8086 requires
    a) an external crystal, whose frequency is equal to the internal frequency of 8086, to be connected to its inputs
    b) an external crystal, whose frequency is twice the internal frequency of 8086, to be connected to its inputs
    c) a clock generated by an external clock generator to be fed
    d) can't say

41. If the operating point of an NPN transistor amplifier is selected in saturation region, it is likely to result in
    a) thermal runaway of transistor
    b) clipping of output in the positive half of the input signal
    c) Need for high dc collector supply
    d) Clipping of output in the negative half of the input signal

42. The MOSFET shown in figure is



    a) common source type
    b) common gate type
    c) source follower type
    d) none of the above

43. When a BJT is turned from ON to OFF, the transistor comes to the OFF state,

a) as soon as the input signal is removed
b) as soon as the power supply is switched off
c) after the excess charge stored in the base region is removed
d) After the input signal is removed

44. When reverse voltage of a diode increases, the capacitance
a) decreases
b) remains same
c) increases
d) has more bandwidth

45. The stack operates as
a) FIFO
b) LIFO
c) LILO
d) None of the above

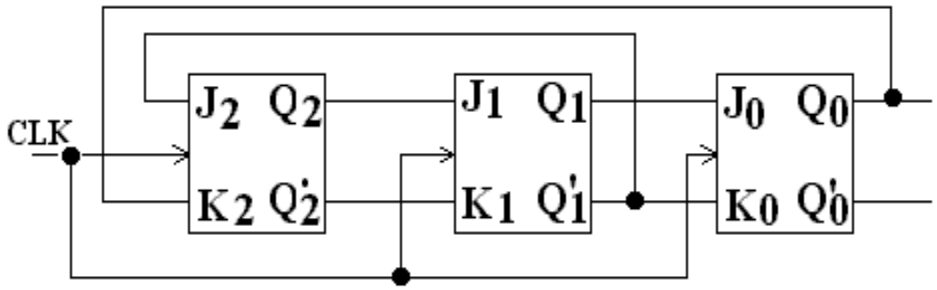
46. The term 'Baud rate' refers to the rate at which
a) parallel data transmission takes place
b) microprocessor operates
c) parallel data can be converted into serial format
d) serial data transmission takes place

47. Most of the digital computers do not have floating point hardware because
a) Floating point hardware is costly
b) It is slower than software
c) It is not possible to perform floating point arithmetic by hardware
d) For no specific reason

48. A carry look ahead adder is frequently used because, it
a) is faster
b) is more accurate
c) Uses fewer gates
d) Costs less

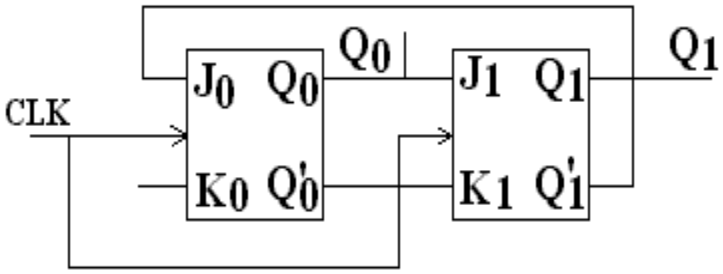
49. In a microcomputer, wait states are used to
a) Make the processor wait during a DMA operation
b) Make a processor wait during an interrupt operation
c) Make a processor wait during a power shutdown
d) Interface slow peripherals to the processor

50. In the following divide by N counter, if initially Q2Q1Q0 = 010; what is the value of N?



a) 4
b) 5
c) 6
d) 7

51. In the following fig. Suppose initially $Q_1Q_2 = 00$. What would be the logic states of Q_1 & Q_2 immediately after 999th clock pulse?



a) Q1Q2 = 00
b) Q1Q2 = 01
c) Q1Q2 = 10
d) Q1Q2 = 11

52. How does one construct a full adder from half adders?
a) One half adder = Full adder
b) Two half adders = Full adder
c) Two half adders and an OR gate = Full adder
d) Two half adders and an AND gate = Full adder

53. To set the ZERO flag in microprocessor which of the following gate is suitable
a) AND gate
b) NAND gate
c) OR gate
d) NOR gate

54. 2's complement of 00010110.01100000 is

a) 11101001.10011110

b) 11101001.10100000

c) 11101010.10100000

d) 11101010.10011110

55. Due to TRI-STATE logic

a) outputs of TTL totem-pole gate can be wired together

b) Propagation delay of gate decreases

c) Fan-in increases

d) None of the above

56. Dynamic RAM can be fabricated using

a) MOS technology

b) TTL

c) ECL

d) I^2L

57. Fusible link is associated with

a) RAM

b) ROM

c) PROM

d) EPROM

58. Twos complement of '-8' is

a) 0111

b) 1000

c) 01000

d) 101000

59. The supply voltage permissible for CMOS device is

a) +5V

b) –5V

c) 3 to 15V

d) +/-12V

60. Tristate logic is used for

a) improving the figure of merit

b) increasing the fan-out

c) bus oriented system

d) improving the speed of operation

61. The number of logic gates required to realize the logical expression Y=AB' is
    a) 1 if AND gate is used
    b) 3 if NAND gates alone are used
    c) 4 if NOR gates alone are used
    d) 4 if EXOR gates alone are used

62. A five input NAND gate is realized using only n input NAND gates. The minimum number of n input NAND gates required is
    a) 2 if n= 8
    b) 4 if n=2
    c) 3 if n= 4
    d) 4 if n=4

63. The logic family used for low power applications is
    a) ECL
    b) TTL
    c) CMOS
    d) NMOS

64. A 4 bit BCD to Excess-3 converter can be realized using four 2^n to 1 multiplexes. The minimum value of n is equal to
    a) 2
    b) 4
    c) 3
    d) 8

65. A 3 to 8 decoder can be implemented using alternate devices as follows
    a) Using three 2 to 4 decoders and an inverter
    b) Using one 1 to 8 demultiplexer
    c) Using one 1 to 16 multiplexer
    d) Using two 2 to 4 decoders

66. The amount of time it takes for the output of a gate or a F/F to change state is denoted as
    a) Hold time
    b) Set up time
    c) Access time
    d) Propagation delay time

67. The decimal equivalent of the hexadecimal number 85 is
    a) 133
    b) 132
    c) 125
    d) 144

68. The no. of chips required to realize 8K x 8 RAM using 8K x 1 RAM is
    a) 4
    b) 8
    c) 16
    d) 2

69. Which of the following gate can be used as an inverter?
    a) AND
    b) OR
    c) EX-OR
    d) None of the above

70. The R-S latch is sensitive to which edge of the clock?
    a) Zero to one transition
    b) One to zero transition
    c) Is not edge sensitive.
    d) The zero to high-impedance transition

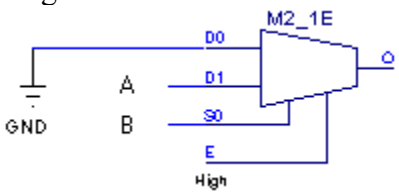
71. Which one can be used as parallel to series converter?
    a) Decoder
    b) Digital counter
    c) Multiplexer.
    d) Demultiplexer

72. Which of the following devices is not a programmable logic device?
    a) AND-OR-INVERT
    b) PAL
    c) ROM
    d) PLA

73. If the AND array is fixed (constrained) and the OR array is programmable, this programmable logic device is classified as a:
    a) EPROM
    b) PAL
    c) PLA.
    d) Flux capacitor

74. What is the maximum memory addressing capacity of given memory map-
0F800H – 0FFFFH
a) 4096
b) 1024
c) 3078
d) 2048

75. In following circuit; if A & B are two inputs and O is output then the circuit diagram is



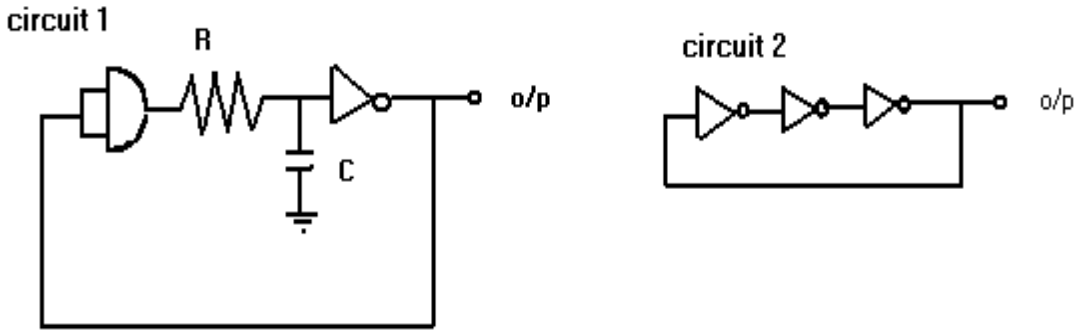
a) OR gate
b) AND gate

c) NAND gate
d) NOR gate

76. Enhancement MOSFET is a normally
a) OFF device
b) ON device
c) Gate current operated device
d) None

77. A R-S Latch is
a) combinational circuit
b) synchronous sequential circuit
c) one bit memory element
d) One clock delay element

78. The circuits given below are



a) Astable – Astable M.V.
b) Astable – Monostable M.V.
c) Monostable – Monostable M.V.
d) Monostable – Astable M.V.

79. Determine the number of flip-flops needed to construct a shift register capable of storing decimal numbers up to 32.
a) 5
b) 6
c) 8
d) 32

80. If we expand the two-variable function AB' wrt B we get ... ?

a) two minterms

b) a canonical form
c) 1

d) none of these

81. The initial state of MOD-16 down counter is 0110. What state will it be after 37 clock pulses?
a) Indeterminate
b) 0110
c) 0101
d) 0001

82. An alternate gate of AND gate is
a) Bubbled I/P OR gate
b) Bubbled I/P NAND gate
c) Bubbled I/P NOR gate
d) Bubbled I/P X-OR gate

83. Each cell of SRAM contains¶
a) 4 MOS transistors and 2 capacitors
b) 6 MOS transistors
c) 2 MOS transistors and 4 capacitors
d) 1 MOS transistor and 1 capacitor

84. The sum of 5 consecutive integers is 35. How many of the five consecutive integers are prime numbers ?
a) 0
b) 1
c) 2.
d) 3

85. What percentage of the number from 1 to 50 have squares that end in the digit 1?
a) 1
b) 5
c) 11
d) 20.

86. Which of the following numbers is the least common multiple of the numbers 2, 3, 4 and 5
a) 12
b) 24

c) 30
d) 60

87. Microprocessor based system can perform many different functions because
a) Its operation is controlled by software.
b) It is digital system
c) It uses a RAM
d) It can be controlled by an input and output devices

88. What gate is used to calculate the partial product when multiplying?
a) XOR
b) AND
c) OR
d) INVERT

89. A microprocessor without the interrupt facility
a) is not useful for a process control system
b) can not be used for DMA operation
c) can not be interfaced with any I/O device
d) will operate at increased speed

90. A TTL NOR gate uses diode logic.
a) True
b) False
c) can't say
d) question is not well formatted

91. The correct sequence in regard to access time of storage devices is
a) floppy disk,hard disk,mag. Tape,RAM
b) RAM,hard disk,mag. Tapc,floppy disk
c) RAM, floppy disk,hard disk,mag. Tape
d) RAM,hard disk,floppy disk,mag.tape

92. The unused input of CMOS circuits should be most appropriately
a) tied to Vcc only
b) tied to GND only
c) either of the above
d) left open

93. An 'AND' gate will function as 'OR' if
a) all the inputs to the gate are '1'
b) all the inputs are "0"
c) either of the input is "1"
d) all the inputs & outputs are complement

94. The resolution of a 4-bit counting ADC is 0.5 volts. For an analog input of-6.6 volts, the digital output of the ADC will be
a) 1011
b) 1101
c) 1100
d) 1110

95. 2's complement representation (signed magnitude) of a 16 bit number is FFFFH. Its magnitude in decimal representation is
a) 0
b) 1
c) 32,767
d) 65,535

96. Memory devices that use electronic latching circuits are called _____.
a) RAM
b) flip-flops
c) magnetic tape
d) DRAM

97. Parity generation and checking is used to detect _____.
a) which of two numbers is greater
b) errors in binary data transmission
c) errors in arithmetic in computers
d) when a binary counter counts incorrectly

98. If an input is activated by a signal transition, it is _____.
a) hair-triggered
b) line-triggered
c) pulse-triggered
d) edge-triggered

99. Pulse-triggered flip-flops are also called _____ FF's.
a) master-slave
b) postponed
c) level
d) edge

100. A retriggerable one-shot has a pulse width of 10mS; 3mS after being triggered, another trigger pulse is applied. The resulting output pulse will be _____ mS.
a. 3
b. 7
c. 10
d. 13

101. FC48 - AB91 = _____.
a. 177
b. 5057
c. 50B7
d. 5077

102. _____ is about twice as fast as P-MOS.
a. CMOS
b. DMOS
c. MOD
d. N-MOS

103. _____is ideally suited for applications using battery power or battery backup power.
a. MOS
b. P-MOS
c. N-MOS
d. CMOS

104. The time it takes for an input signal to pass through internal circuitry and generate the appropriate output effect is known as _____.
a. fan-out
b. propagation delay
c. rise time
d. fall time

105. One advantage that MOSFET transistors have over bipolar transistors is ____.
a. high input impedance
b. higher switching speed
c. low input impedance
d. reduced propagation delay

106. A(n) _____ is a logic circuit that accepts a set of inputs which represents a binary number, and activates only the output that corresponds to that input number.
a. decoder
b. LED
c. encoder
d. demultiplexer

107. A(n) _____ converts an analog input to a digital output.
a. DC
b. DAC
c. flash converter
d. bipolar converter

108. A major application for DSP is in _____ and _____ of analog signals.
a. sending, receiving

b. digitizing, weighting
c. filtering, conditioning
d. leveling, translating